

## CLINICAL PROBLEM-SOLVING

Caren G. Solomon, M.D., M.P.H., *Editor*

## Turning Purple with Pain

Thomas W. Fredrick, M.D., Manuel B. Braga Neto, M.D., Ph.D.,  
Daniel O. Johnsrud, M.D., Michael Camilleri, M.D., and Victor G. Chedid, M.D.*In this Journal feature, information about a real patient is presented in stages (boldface type) to an expert clinician, who responds to the information by sharing relevant background and reasoning with the reader (regular type). The authors' commentary follows.*

**A 32-year-old woman with a 10-year history of abdominal pain presented to the gastroenterology clinic for evaluation of abdominal pain. She reported episodes of severe, diffuse abdominal pain, which she rated at 10 on a scale of 0 to 10, with 10 representing the worst pain imaginable. These episodes occurred every month or two and usually lasted for several days. They were unrelated to her menstrual cycles and occurred without an apparent precipitant. Between episodes, she reported having mild, cramping midabdominal pain almost every day, with the pain fluctuating in severity throughout the day. The pain was not associated with food intake but was aggravated by running. She had been hospitalized three times for episodes of severe pain.**

In a young woman who presents with episodic acute-on-chronic abdominal pain, the differential diagnosis is broad. Possible causes include chronic constipation, inflammatory bowel disease (IBD), and irritable bowel syndrome (IBS). IBD is often associated with systemic symptoms such as fatigue, weight loss, diarrhea, and arthralgia. A detailed history of her bowel function should be obtained.

The episodes of severe abdominal pain were followed by 3 to 4 days of constipation, without excessive straining, manual digitation to defecate, blood in the stool, weight loss, fevers, or changes in appetite. The patient reported taking over-the-counter laxatives when pain occurred, after which she had loose stools and partial pain relief. Her bowel function was normal between the episodes of pain. Her only medications were an etonogestrel subdermal implant for contraception, which had been placed 3 years before presentation and resulted in amenorrhea, cetirizine for seasonal allergies, and valacyclovir for herpes simplex prophylaxis. Before receiving the implant for contraception, she had taken combined oral contraceptives for at least 5 years. Her medical and surgical history were notable for the removal of an ovarian cyst by means of a midline infraumbilical laparotomy at 16 years of age and a diagnosis of fibromyalgia at 15 years of age. She had never smoked, and she ran marathons. She drank alcohol socially. She reported a history of childhood sexual abuse. She had tried multiple diets for weight loss, including ketogenic diets. There was no family history of IBD or abdominal pain similar to what the patient reported.

The absence of diarrhea, blood in stools, weight loss, and other systemic symptoms argue against IBD. The severe episodic symptoms raise the possibility of angioedema or systemic mastocytosis. These conditions are associated with atopy and other systemic manifestations, such as rashes and urticaria. The patient's history of childhood sexual abuse may predispose her to chronic abdominal or pelvic

From the Department of Internal Medicine (T.W.F.) and the Division of Gastroenterology and Hepatology, Department of Medicine (M.B.B.N., M.C., V.G.C.), Mayo Clinic, Rochester, MN; and Aurora Cardiovascular and Thoracic Services, Aurora Sinai Medical Center and Aurora St. Luke's Medical Center, University of Wisconsin School of Medicine and Public Health, Milwaukee (D.O.J.). Address reprint requests to Dr. Chedid at the Division of Gastroenterology and Hepatology, Department of Medicine, Mayo Clinic, Rochester, MN 55905 or at [chedid.victor@mayo.edu](mailto:chedid.victor@mayo.edu).

N Engl J Med 2021;385:549-54.

DOI: 10.1056/NEJMcps2105278

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pain and functional gastrointestinal disorders. Her intermittent constipation could indicate IBS with constipation, functional constipation, pelvic-floor dysfunction, or a secondary cause of constipation, such as hypothyroidism or hypercalcemia. It may also be related to medication. However, the intermittent nature of the constipation makes hypothyroidism, hypercalcemia, and pelvic-floor dysfunction less likely. Some of the medications she takes, such as cetirizine, have been associated with constipation but are unlikely to explain her symptoms.

Given the patient's history of constipation, a digital rectal examination is indicated. Findings of high resting anal sphincter tone, reduced perineal descent (<2 cm), and paradoxical contraction of the pelvic floor on palpation with the examining finger during straining would suggest pelvic-floor dysfunction as a cause of constipation. Physical examination for her abdominal pain should include assessment of Carnett's sign — a localized area of abdominal tenderness when the patient tenses the abdominal muscles by flexing the neck — which indicates musculoskeletal abdominal pain. However, the episodic nature of her symptoms argues against both pelvic-floor dysfunction and musculoskeletal abdominal pain.

Although the patient's symptoms are not cause for alarm (i.e., they do not include vomiting, fevers, or gastrointestinal bleeding), the severity of the acute exacerbations of pain is worrisome and requires further evaluation. Records related to her hospitalizations should be obtained in order to review the results of previous evaluations.

**During previous hospitalizations, the patient's abdominal pain had been managed conservatively, with bowel rest and opioids for pain. Investigations included upper gastrointestinal endoscopy (duodenal-biopsy specimens were negative for celiac disease) and colonoscopy with examination of the terminal ileum (findings were normal). Several abdominal and pelvic computed tomographic (CT) scans had been obtained to determine the cause of acute pain and were either normal or showed nonspecific findings. During her most recent hospitalization, which occurred 3 days before the current outpatient presentation, laboratory testing revealed a low serum sodium level (with a nadir of 124 mmol per liter), which was attributed to hypovolemia and corrected with the intravenous administration of normal saline.**

**Other electrolyte levels were normal. During this hospitalization, the patient was noted to appear to be transiently "out of it," a condition that was thought to be related to the use of excessive opioid medication for pain. Concerns for opioid-seeking behavior were raised by providers. Psychiatric evaluation led to a diagnosis of conversion disorder.**

During the current outpatient visit, the patient was alert and oriented to time and place and was not in distress. Her body-mass index (the weight in kilograms divided by the square of the height in meters) was 19.5, and she weighed 55.7 kg. She had abdominal tenderness to deep palpation, with a negative Carnett's sign and normal results on digital rectal examination, without findings that would suggest pelvic-floor dysfunction. A complete blood count was normal, as were levels of creatinine, blood urea nitrogen, ferritin, thyrotropin, and fasting morning cortisol. The aspartate aminotransferase level was 48 U per liter (normal range, 8 to 43), and the alanine aminotransferase level was 53 U per liter (normal range, 7 to 45). The tryptase level was 9.5 ng per milliliter (normal range, <11.5), the C1 esterase inhibitor level 34 mg per deciliter (normal range, 19 to 37), and the tissue transglutaminase (tTG) antibody level less than 1.2 U per milliliter (normal range, <4). IgA levels were normal. Nortriptyline was started at a dose of 10 mg per day as a central neuromodulator to relieve presumed functional abdominal pain.

Adrenal insufficiency can cause abdominal pain, constipation, and hyponatremia, but the normal morning cortisol level is inconsistent with this diagnosis. The normal levels of C1 esterase inhibitor and tryptase levels argue against angioedema and systemic mastocytosis, respectively. Together with the history of normal findings on biopsy of the small bowel, the normal tTG-IgA level (with normal IgA levels) does not support a diagnosis of celiac disease. Functional abdominal pain is frequently diagnosed in patients for whom an extensive diagnostic workup has been unrevealing, and nortriptyline, along with dietary modification and cognitive behavioral therapy, may be helpful if that is the cause of the patient's symptoms. Her transient hyponatremia could be explained by hypovolemia associated with decreased oral intake due to abdominal pain.

**The patient returned to the emergency department 2 months later with a 3-day history of severe**

abdominal pain and no bowel movement or flatus passed for 1 day. Her pulse rate was 107 beats per minute, blood pressure 113/81 mm Hg, respiratory rate 14 breaths per minute, and oxygen saturation 100% while breathing ambient air. She appeared uncomfortable and anxious but was oriented to time and place, spoke clearly, and responded appropriately to questions. Her mucous membranes were dry, and she had tachycardia with regular rhythm. Her abdomen was diffusely tender; bowel sounds were high pitched. Results of a digital rectal examination were normal, with no evidence of fecal impaction.

Although the patient had had previous CT scans that were either unremarkable or inconclusive, her current severe pain, the lack of flatus or bowel movements for 1 day, and the findings on the abdominal examination raise concern for a bowel obstruction. Repeat abdominal imaging should be performed.

A complete blood count and levels of glucose, creatinine, sodium, total bilirubin, alkaline phosphatase, lactate, and lipase were within normal limits. The result of a urinary pregnancy test was negative, and no nitrites or leukocytes were detected on urinalysis. Contrast enhanced CT of the abdomen and pelvis revealed only a large stool burden.

Intravenous hydration (with crystalloids) and pain medication (hydromorphone) were initiated in the emergency department, and the pain abated and the tachycardia resolved. The patient was admitted to the hospital, and treatment with tramadol and acetaminophen was started, replacing hydromorphone. The constipation persisted despite treatment with a bisacodyl suppository, magnesium citrate, oral senna and docusate, lactulose, polyethylene glycol, sodium phosphate, and a mineral oil enema. Anxiety was treated with hydroxyzine and lorazepam as needed. On hospital days 2 and 3, the patient continued to have severe abdominal pain. On hospital day 3, her serum sodium level fell from 136 mmol per liter on admission to 128 mmol per liter. Given the presumption of hypovolemic hyponatremia, 500 ml of normal saline was administered intravenously, but a repeat measure of sodium levels obtained later that day showed a further decline to 125 mmol per liter. The serum osmolality was 255 mOsm per kilogram, the urine osmolality 413 mOsm per kilogram, and the urine sodium level 153 mmol

per liter, all of which suggested the syndrome of inappropriate antidiuretic hormone secretion (SIADH). Water restriction was initiated, and tramadol was discontinued given its rare association with hyponatremia. The serum thyrotropin level was 2.4 mIU per liter (normal range, 0.3 to 4.2), and the morning serum cortisol level was 24  $\mu$ g per deciliter (normal range, 7 to 25).

Laboratory and imaging studies have ruled out bowel obstruction, pancreaticobiliary disease, mesenteric ischemia, pregnancy, and urinary tract infection. Hyponatremia in the setting of clinical evidence of euolemia is most consistent with SIADH. Abdominal pain can trigger increased antidiuretic hormone secretion and, when combined with poor oral intake, can cause hyponatremia. Although a similar scenario may result from adrenal insufficiency or hypothyroidism, the patient's recent thyrotropin and morning cortisol levels were reportedly normal. Other causes include anorexia nervosa, diabetes mellitus, psychogenic polydipsia, and, in rare cases, porphyrias. The patient's borderline low BMI and previous attempts at a restrictive diet raise concern for anorexia nervosa. Diabetes mellitus is unlikely given her normal fasting glucose level. Psychogenic polydipsia is also unlikely in a monitored hospital setting. Acute intermittent porphyria is high on the differential diagnosis given her episodic severe abdominal pain, hyponatremia, changes in mental status on a previous admission, and age (young and middle-aged women are most commonly affected). Treatment of hyponatremia is based on correction of the underlying problem and fluid restriction to reduce free water intake and, in severe hyponatremia, initiate hypertonic saline. However, the degree of hyponatremia in this patient does not warrant hypertonic saline treatment.

On hospital day 4, the patient remained constipated, and a water-soluble enema administered with contrast material revealed fecal impaction in the transverse colon. The serum sodium level fell to 122 mmol per liter. By hospital day 6, her constipation and abdominal pain had gradually abated with laxative management, and the hyponatremia had abated with fluid restriction. Owing to the constellation of recurrent severe abdominal pain, hyponatremia, and transient neuropsychiatric manifestations, a urine porphyrin panel was obtained and revealed elevated levels of porphobi-

linogen (991.8  $\mu\text{mol}$  per liter; upper limit of the normal range, <1.3) and  $\delta$ -aminolevulinic acid (318 nmol per milliliter; upper limit of the normal range, <15). Subsequent genetic testing identified a heterozygous frameshift alteration in *HMBS* (g.118960947dup, from genome reference GRCh38), predicting amino acid change p.F158VfsTer52 and hydroxymethylbilane synthase deficiency and consistent with a diagnosis of acute intermittent porphyria. After the results of urinalysis were returned, the nurses taking care of the patient noted that her urine had appeared to be “purple colored” a few days earlier.

On further questioning once the diagnosis was made, the patient reported no family history of porphyria. She noted that previous hospitalizations had occurred when she was following a ketogenic diet. The medical team advised her to avoid such diets in the future.

On outpatient follow-up 2 months after diagnosis, the patient reported substantial abatement of her abdominal pain and constipation. She also noted that she had been able to return to work as a graphic artist.

#### COMMENTARY

This patient presented with a history of episodes of severe, acute abdominal pain and constipation, superimposed on chronic pain, that had been unexplained despite extensive evaluations. Ultimately, the associated psychiatric symptoms during an earlier hospitalization and the identification of hyponatremia during two previous hospitalizations raised suspicion that acute intermittent porphyria was the cause of her pain. The documentation of elevated urinary porphyrin levels confirmed the diagnosis.

Porphyrias are a group of metabolic disorders that result from defects in enzymes within the heme biosynthetic pathway.<sup>1</sup> The most common are porphyria cutanea tarda, acute intermittent porphyria, and erythropoietic protoporphyria, in that order.<sup>1</sup> Acute intermittent porphyria results from mutations in *HMBS*, the gene encoding hydroxymethylbilane synthase (see Fig. 1), which plays an integral role in heme synthesis.<sup>1,2</sup> The disease manifests as acute attacks in which the heme pool in the liver is used up, leading to a buildup of  $\delta$ -aminolevulinic acid and porphobilinogen (PBG).<sup>1,2</sup>

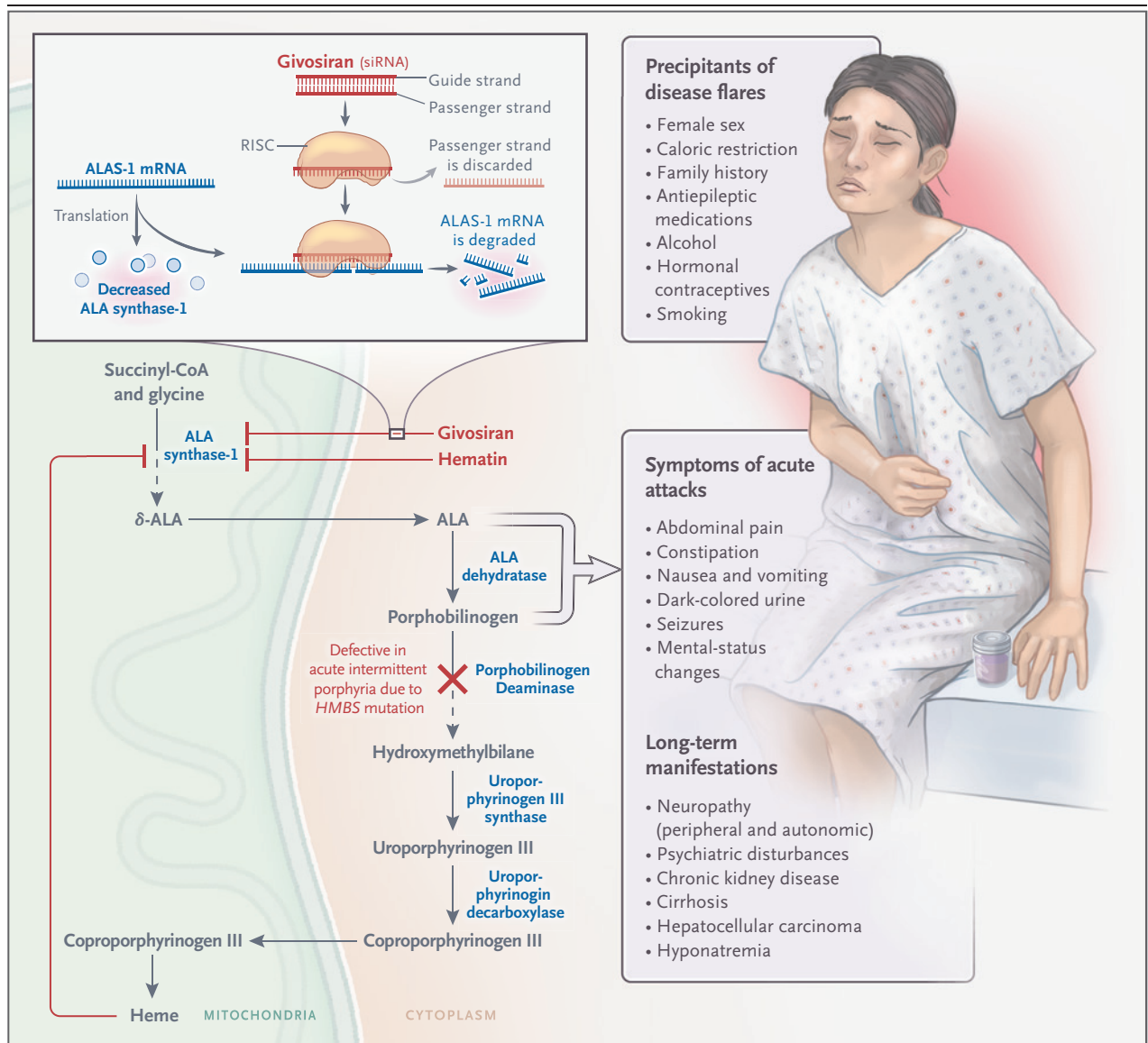
Acute intermittent porphyria is a rare disease, with an estimated prevalence of 5.4 cases per

million according to a survey conducted in European countries,<sup>3</sup> although this may be an underestimate. The disease has incomplete penetrance (approximately 10 to 20% in genetically affected persons). The typical patient is a woman between the ages of 18 and 45 years.<sup>1,4</sup> In a case series of 108 persons with acute intermittent porphyria, common symptoms included abdominal pain (74%), nausea and vomiting (73%), constipation (60%), and anxiety and depression (55%). Although symptoms were intermittent and acute in 55% of the patients, approximately 18% reported nearly constant symptoms, including chronic abdominal pain, as in our patient.<sup>5</sup> Coexisting conditions include hypertension (43%), peripheral neuropathy (43%), chronic kidney disease (29%), psychiatric disorders (22%), palpitations (19%), seizures (9%), cirrhosis (2%), and hepatocellular carcinoma (1%).<sup>5</sup> Elevated levels of aminolevulinic acid accumulate in the liver, the central and peripheral nervous system, and the kidneys and may account for several of these coexisting conditions.<sup>6</sup> Patients with peripheral neuropathies often present with weakness that begins in the legs and ascends.<sup>5,7</sup> In addition, autonomic neuropathy may develop, resulting in cardiac arrhythmias, dysautonomia, and gastrointestinal manifestations.<sup>6</sup> In a cohort study from Sweden, researchers reported a significantly increased risk of diagnoses of schizophrenia and bipolar disorder among persons with acute intermittent porphyria.<sup>8</sup> Given the challenges of diagnosing acute intermittent porphyria, patients may receive a misdiagnosis with primary psychiatric conditions.

Owing to the reported association between acute intermittent porphyria and hepatocellular carcinoma, experts recommend annual screening with hepatic ultrasonography as well as screening for alpha-fetoprotein starting at 50 years of age in patients with any form of acute porphyria.<sup>1,5,9</sup> Although data from randomized trials are lacking to support the benefit of this approach, biennial screening in a Swedish cohort was associated with an improvement in survival of 3 to 5 years as compared with no or infrequent screening; however, lead-time bias (whereby seemingly longer survival is attributable to earlier diagnosis) may explain these findings.<sup>9</sup>

Flares of acute intermittent porphyria can be precipitated by the ingestion of alcohol, infections, low caloric intake, elevated levels of reproductive hormones, and medications, particularly





**Figure 1. Pathophysiology of Acute Intermittent Porphyria.**

Acute intermittent porphyria is caused by heterozygous mutations in *HMBS*, the gene encoding hydroxymethylbilane synthase, which results in a deficiency of porphobilinogen deaminase, an enzyme that plays an integral role in heme synthesis. When a patient is exposed to certain triggers, such as caloric restriction, alcohol, or causative medications,  $\delta$ -aminolevulinic acid synthase (ALAS) activity ramps up in an attempt to increase heme synthesis. This leads to a buildup of  $\delta$ -aminolevulinic acid (ALA) and porphobilinogen, which causes systemic symptoms of pain and neuropathy as well as psychiatric disturbances. Treatment relies on the infusion of hematin, which down-regulates ALA through negative feedback, or givosiran, which inhibits the synthesis of ALAS. RISC denotes RNA-induced slicing complex. CoA denotes coenzyme A, mRNA messenger RNA, and siRNA small, interfering RNA.

antiseizure medications.<sup>1,10</sup> Our patient reported a history of carbohydrate restriction in keeping with ketogenic diets as well as the use of hormonal contraception, both of which may have contributed to her frequent episodes of severe pain.

Common but nonspecific laboratory findings associated with the onset of an acute intermittent porphyria attack include hyponatremia, hypo-

magnesemia, mild elevations of aminotransferase levels, and mild leukocytosis.<sup>1,5,7,10</sup> Hyponatremia, as observed in our patient, has been documented in 25 to 60% of acute attacks and is attributed to elevated levels of antidiuretic hormone.<sup>11,12</sup>

In patients with acute intermittent porphyria, the urine can appear red or brown and darkens on exposure to oxygen, light, or heat, because

urinary porphyrins undergo autoconversion of PBG into porphobilin.<sup>1,2</sup> The purple urine noted near the time of diagnosis in this patient reflects the very high urinary PBG level. Elevated PBG levels (relative to the urinary creatinine level) obtained in a random urine sample — which must be protected from light — are consistent with the diagnosis, and a normal value obtained during an acute flare rules out acute porphyria.<sup>13</sup> During a flare, PBG levels can be as high as 20 to 300 mg per gram of creatinine (normal range, 0 to 2 mg). Although a urinary creatinine level was not available in our patient, her urinary PBG level was sufficiently high to support the diagnosis. Urine samples obtained several days after a flare may show persistent PBG elevation.<sup>13</sup> The diagnosis should be confirmed by genetic testing in which a mutation in the gene encoding hydroxymethylbilane synthase is detected.<sup>4,13</sup>

Treatment of flares starts with intravenous administration of dextrose, which inhibits transcription of the gene encoding transhepatic 5'-aminolevulinic synthase 1 (ALAS1).<sup>1,13</sup> A high-carbohydrate diet has also been suggested in case reports,<sup>5,13</sup> and parenteral opioids are sometimes used in managing pain. However, up to 50% of patients may not report substantive improvement with opioids, and frequent or long-term use of opioids can have adverse effects, including the development of dependence.<sup>5</sup> Hematin, administered intravenously, is also recommended; this treatment effectively bypasses the defective *HMB* gene. Uncontrolled studies have suggested bene-

fit, which may take up to 3 days; one small, placebo-controlled trial did not show clear evidence of benefit regarding symptoms of intermittent porphyria, but the trial was underpowered.<sup>14</sup> Givosiran, an RNA interference therapy targeting *ALAS1* messenger RNA, was recently approved for the prevention of porphyria flares on the basis of benefit reported in a phase 3 randomized trial in which givosiran significantly reduced the mean annual attack rate from 12.5 to 3.2 as compared with placebo.<sup>15</sup> Patients should be counseled to avoid triggers, including alcoholic beverages, calorie restriction, and predisposing medications (including hormonal contraception). Premenopausal women may benefit from hormonal suppression.<sup>13</sup> The prognosis is good if the disease is identified and measures are taken to minimize the risk of flares and to effectively treat those that occur.

This case illustrates the difficulty in diagnosing rare diseases such as acute intermittent porphyria when patients present with nonspecific manifestations. Our patient had several classic symptoms and signs of acute intermittent porphyria, including severe, episodic abdominal pain and constipation, transient psychiatric manifestations, hyponatremia, and discolored urine. Earlier recognition of these symptoms and a higher level of suspicion regarding acute intermittent porphyria could have led to more prompt diagnosis and effective management.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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